

Beyond a One-Size-Fits-All Approach: Towards Personalizing Multi-device Setups in Crowdwork

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ABSTRACT

Non-workstation devices provide mobility in the sense that they enable work and play in less constrained environments and configurations. Workers have different needs and preferences in terms of work practices that impact work-life balance. One way to increase mobility is to allow for more choices in both workstation and non-workstation devices. This paper investigates whether workers' preferences for using different devices are distinct. To understand whether these differences in preference exist, we use data from an exploratory qualitative survey of 150 crowdworkers from Amazon Mechanical Turk. We first identify the aspects that influence crowdworkers' preferences in using non-workstation devices. Next, our thematic analysis of open-ended data indicates that different workers have different preferences for multi-device configuration. Finally, this leads to a discussion of how people have different preferences which would require different non-workstation devices.

CCS CONCEPTS

• **Human-centered computing** → **Qualitative studies in Ubiquitous computing**.

KEYWORDS

Crowdwork, Personalizing multi-device, Human Intelligence Tasks

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1 INTRODUCTION

With the advancement of availability and technical capabilities there has been a growth in the types of non-workstation devices

which includes smartphones, tablets, smart speakers, and smartwatches. *Mobility* is a key attribute of non-workstation devices, which we define it as the ability to work in a non-workstation environment. Enabling work to be done in less constrained environments and configurations, mobility is an important aspect of work life, but its importance is determined by an individual's preferences for work life. There are primarily two categories of individuals based on their work-life preferences or choices: (1) those who do not have a clear demarcation line between work and personal life; and (2) people who maintain a strong distinction between their professional and personal lives. For example, Digital nomads, Work warriors are type of people who prefer integration at ease with blurring the lines between work and non-work life [6, 8]. In contrast, Separators and Family Guardians [1, 4, 7, 8] have been identified as a group of people who set a distinct boundary between work life and personal life. Mobility can be achieved through different combinations of workstation based as well as non-workstation devices in crowdwork such as using a desktop and smartphone [2], using a desktop, smartphone and tablet [5], or a desktop, smartphone, smart speaker combination [3]. Therefore we would like to investigate whether crowdworkers have preferences in using combinations of multiple devices for their work.

2 DATA COLLECTION AND ANALYSIS

We begin the online survey by asking about participants' personal demographics (i.e., age, gender, education) and their work experience on Amazon Mechanical Turk (e.g., completed HITs, current work hours, HIT approval rate). For this work in progress, we report specifically on five results from five of the survey's questions: We asked participants to consider a scenario in which they have a magic wand that allows them to change the work on Amazon MTurk in whatever way they wished. Following is one example of open-ended questions which were asked for tablets to the participants: "How would you use the magic wand to make managing and performing HITs better for a tablet? What would you change? Why?" We asked the same question for the remaining four types of devices: desktops, smartphones, smart speakers, and smartwatches. We recruited a total of 150 participants for the study (Male = 97, Female = 53) by deploying the survey to Amazon MTurk. 148 participants (98.7%) are at least 25 years older and leaving 2 participants (1.3%) belonging to the group of 18-24. In terms of education, 105 participants (70%) had a Bachelor's degree or higher. 119 (79.3%) participants has stated

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that they have worked on the Amazon MTurk platform for 2 or more years. 57 participants (38%) mentioned that they work 10 to 20 hours per week on MTurk and 32 participants (21.3%) indicated that they work 30 or more hours each week. The median of total HITs was 24,170 ($\sigma=205,826$), and the median approval rating was 99.5% ($\sigma=45.34$). To guarantee participants' data was both trustworthy and driven by experience with crowdwork, we implemented a HIT qualification that required individuals to have completed at least 10K HITs and have a minimum acceptance rate of 98.0%. According to recent research, crowdworkers on Amazon Mechanical Turk may multitask when the pay is insufficient. As a result, we decided to pay participants 5.00 USD to ensure they are appropriately rewarded and feel extremely comfortable devoting their whole attention to our HIT [8].

All the open-ended questions related to magic wand are evaluated with a bottom-up approach by developing codes to find themes. Our overall objective with bottom-up thematic coding is to naturally monitor how participants like to use various devices in their work.

3 PRELIMINARY RESULTS

Our qualitative results first outline the factors that affect different needs for different type of devices to be involved in crowdsourcing workflows. Participants mention about different types of factors needed for different kinds of non-workstation devices. For example, touch screen capable HITs for tablets, having the ability to use voice commands in order to complete or review a HIT on smart speakers, using one click for completing HITs on smartwatches.

"I stopped using my tablet because I couldn't always use touch screen, so probably making more HITS touch screen capable." (P59)

"...I would make it a one click acceptance...Also, simplify scrolling process on smartwatch. This would make it easier to catch and complete HITs." (P107)

"The ability to vocally review HITs. If I dislike [the HIT] smart speaker speaks out why [I dislike it] or get a voice prompt when completing a HIT." (P168)

Our thematic analysis process identified the following three themes which shows the different preferences for using multi-devices for crowdwork: (1) *Multi-device for HIT Management*: refers to the set of participants who would like to manage HITs using multiple devices. For example, respondents mention about having the ability to run scripts on other devices along with desktops (e.g. smartphones, tablets, smart speakers and smartwatches) to manage HITs. For instance, utilizing scripts find or catch (by scraping or filtering Amazon MTurk) HITs by HIT ID, and auto-accepting HITs based on a specified condition (for example, a certain minimum wage) defined by the crowdworker. They also mention about accepting HITs using voice commands on smart speakers.

"I'd like to see more scripts/tools on smartphone that allow the 'catching' of HITs like desktop." (P102)

(2) *Device Segregator*: represent the respondents who prefer different devices for HIT completion and HIT management. For example, P96 mention about using smartphones and tablets for HIT completion and smart speakers and smartwatches for managing HITs (e.g. catching HITs, receiving notification for accepted HITs). Our result

from online survey reveals that there are more workers like participant 96 who would choose to utilize a certain type or different types of devices for HIT completion vs management.

(3) *General Openness for Multi-devices*: This theme addresses the participants who shows a general openness to use multiple devices for crowdwork. Our analysis of free-form responses shows that some participants are interested to use all different kinds of devices for crowdwork. For example, participants indicate a desire to be able to conveniently execute tasks using multi-devices (e.g. desktop, smartphone, tablet, smart speaker and smartwatch).

"[I would like to] make it easier to accept HITs and work on them or even find good work." (P150)

Overall our results indicate that there are likely differences in the specifics of multi-device setup, depending on crowdworker's preference.

4 CONCLUSION AND SUBSEQUENT STEP

Our overall findings indicate that there is a potential for recognizing workers with different preferences, which would require the use of different non-workstation devices. We believe that it will be useful to build quantitative foundations for identifying workers with diverse preferences. This can be useful for designing intelligent work assistant for crowdworkers, so that the systems can adjust the workflow to be suitable for particular workers based on their preference. Therefore we would like to explore whether it is possible to identify workers based on their work preferences.

REFERENCES

- [1] Blake E Ashforth, Glen E Kreiner, and Mel Fugate. 2000. All in a day's work: Boundaries and micro role transitions. *Academy of Management review* 25, 3 (2000), 472–491.
- [2] Senjuti Dutta, Rhema Linder, Doug Lowe, Richard Rosenbalm, Anastasia Kuzminykh, and Alex C Williams. 2022. Mobilizing crowdwork: A systematic assessment of the mobile usability of hits. In *CHI Conference on Human Factors in Computing Systems*. 1–20.
- [3] Danula Hettiachchi, Senuri Wijenayake, Simo Hosio, Vassilis Kostakos, and Jorge Goncalves. 2020. How Context Influences Cross-Device Task Acceptance in Crowd Work. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, Vol. 8(1). 53–62.
- [4] Ellen Ernst Kossek and Brenda A Lautsch. 2008. *CEO of me: Creating a life that works in the flexible job age*. Pearson Prentice Hall.
- [5] Gemma Newlands and Christoph Lutz. 2021. Crowdwork and the mobile underclass: Barriers to participation in India and the United States. *new media & society* 23, 6 (2021), 1341–1361.
- [6] Ina Reichenberger. 2018. Digital nomads—a quest for holistic freedom in work and leisure. *Annals of Leisure Research* 21, 3 (2018), 364–380.
- [7] Patricia Voydanoff. 1998. Home and work: Negotiating boundaries through everyday life. *Contemporary Sociology* 27, 2 (1998), 153.
- [8] Alex C Williams, Gloria Mark, Kristy Milland, Edward Lank, and Edith Law. 2019. The perpetual work life of crowdworkers: How tooling practices increase fragmentation in crowdwork. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–28.